



Introduction to the Course

Introduction to Machine Learning

Embedded Machine Learning

✓ **Video:** Machine Learning on Embedded Devices
5 min

✓ **Video:** Machine Learning Specific Hardware
13 min

✓ **Video:** Machine Learning Software Frameworks
7 min

✓ **Reading:** Machine Learning on Microcontrollers
5 min

📋 **Practice Quiz:** Embedded Machine Learning
5 questions

Using Edge Impulse to Collect Data

▶ **Video:** Getting Started with Edge Impulse
6 min

▶ **Video:** Data Collection
14 min

📖 **Reading:** Edge Impulse CLI Installation Troubleshooting
10 min

📖 **Reading:** What Makes a Good Dataset
5 min

📋 **Practice Quiz:** New Quiz
5 questions

Feature Extraction

▶ **Video:** Feature Extraction from Motion Data
10 min

▶ **Video:** Feature Selection in Edge Impulse
4 min



Machine Learning on Microcontrollers

While optional, we encourage you to take a look at the following articles to learn more about the topics covered in this lesson:

[TinyML: Making Smart Devices Tinier than Ever](#)

[Signal processing is key to embedded Machine Learning](#). This article is a bit advanced, as it includes topics we have not covered yet. However, I recommend checking it out if you're curious to see how digital signal processing (DSP) complements embedded machine learning.

If you'd like to learn more about AI accelerators, including various processor architectures intended to optimize machine learning algorithms, I highly recommend checking out the following articles:

- [What Makes a Good AI Accelerator](#)
- [How to Make Your Own Deep Learning Accelerator Chip](#)
- [DNN Accelerator Architecture - SIMD or Systolic?](#)

✓ Complete

Go to next item

